

Test Exam Solution

Introduction to Database Systems – Spring 2026

General Information

Instructions:

- You have 4 hours to answer the questions in the following sections. You may have extra time if SAP has granted you special exam conditions.
- You must provide all your answers online, directly in this Exam Quiz.

Negative points: in all multiple-choice questions, wrong answers are assigned negative points so that the total score for all incorrect answers cancels out the total score for all correct answers. That is, the sum of all correct answers is 100%, while the sum of all wrong answers is -100%. For example, if you have 5 options, two are correct and three are wrong, each correct answer is worth 50% and each wrong answer is worth -33.33%. Note that the final grade for the question cannot be negative; the lowest possible grade is zero.

Important: it is your responsibility to ensure that you SUBMIT and FINISH your attempt: if an exam is not properly submitted, we WILL NOT take it into account.

Good luck!

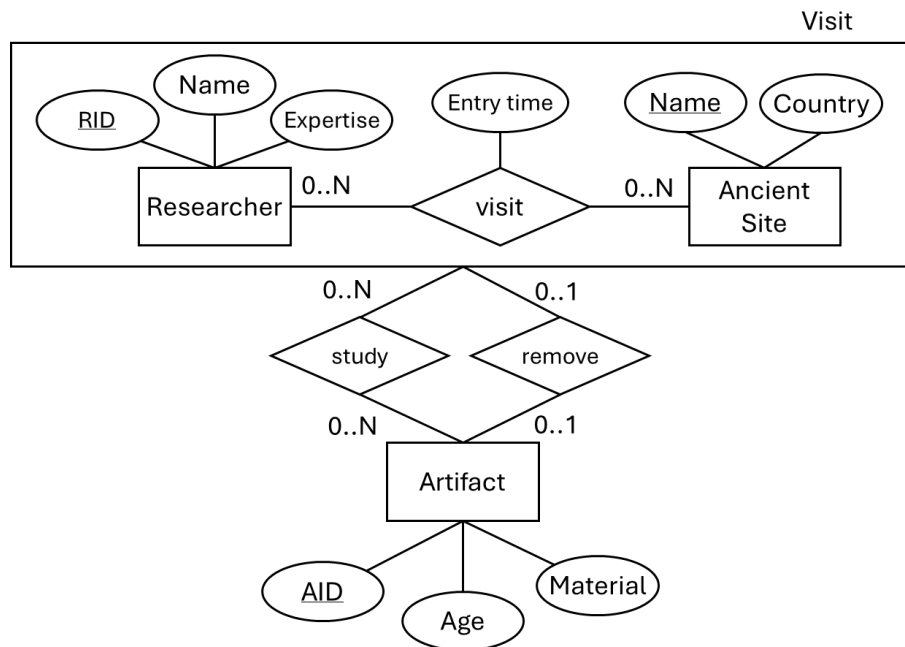
ER Modeling, DDLs, and Normal Forms (25 points)

Question 1 (7 points)

Draw an entity-relationship (ER) diagram for the following use case. The ER diagram must follow the notation presented in the "Principles of Database Management" (PDBM) book, as amended in lectures. Make sure to include entities, relationships, primary keys, cardinalities, and other features of ER models where applicable. Add an ID as a primary key only if no other attribute can serve as a primary key. Adding unnecessary primary keys will result in a loss of points. Export and upload the diagram in PNG or JPG format.

Researchers visiting an ancient site often study multiple artifacts during their visit, although some visits may result in no artifacts being studied. The entry time to the ancient site is always recorded. Ancient sites are identified by their name and are located in a specific country. Researchers are identified by a unique researcher ID and also have a name and a field of expertise. Artifacts are characterized by their estimated

age and material. All artifacts are at least 500 years old. During each visit to an ancient site, a researcher may study many artifacts but may remove at most one artifact for further research purposes. Each artifact can be removed only once.



Question 2 (7 points)

Write SQL Data Definition Language (DDL) statements to create tables for each entity and relationship (if necessary) identified in the ER model from the previous question. Specify appropriate data types for each attribute and include primary keys and foreign keys to enforce the relationships between tables. Additionally, add constraints to enforce as many rules as possible from the problem statement of Question 1.

The SQL DDL statements must be executable in DuckDB. Statements that do not run in DuckDB will result in a loss of points.

```

create table Researcher (
    rid int primary key,
    name string not null,
    expertise string not null
);
    
```

```

create table AncientSite (
    name string primary key,
    country string not null
);
    
```

```
create table Artifact (
    aid int primary key,
    age int not null check (age >= 500),
    material string not null
);
```

```
create table Visit (
    rid int,
    siteName string,
    entryTime datetime,
    artifactRemoved int unique,
    primary key (rid, siteName, entryTime),
    foreign key (rid) references Researcher(rid),
    foreign key (siteName) references AncientSite(name),
    foreign key (artifactRemoved) references Artifact(aid)
);
```

-- artifactRemoved may be NULL if no artifact was removed; it must be unique because each artifact can be removed only once. If there is a separate VisitID, then ensure that (rid, siteName, entryTime) is unique and that none of these fields is NULL.

```
create table Study (
    rid int,
    siteName string,
    entryTime datetime,
    artifactStudied int references Artifact(aid),
    primary key (rid, siteName, entryTime, artifactStudied),
    foreign key (rid, siteName, entryTime) references Visit(rid, siteName, entryTime)
);
```

Question 3 (5 points)

Take one relation from the ancient site database of Question 1 and modify it so that the relation is in second normal form (2NF) but not in third normal form (3NF). You may merge multiple relations from the given database, modify primary keys, and add or remove attributes.

List the relation and the functional dependencies so that the relation is in 2NF but not in 3NF.

Solution:

Add a unique ID VisitID to the Visit relation and make it the primary key. Merge relations Researcher and Visit into ResearcherVisit with VisitID as the primary key (and only candidate key):

ResearcherVisit(VisitID, RID, SiteName, EntryTime, ArtifactRemoved, Name, Expertise)

Functional dependencies:

- VisitID → RID, SiteName, EntryTime, ArtifactRemoved
- RID → Name, Expertise

The relation ResearcherVisit is in 2NF because its only candidate key, VisitID, is a single attribute, so no non-prime attribute can be partially dependent. However, it is not in 3NF because the non-prime attributes Name and Expertise are transitively dependent on VisitID via RID, violating 3NF.

Question 4 (3 points)

Given a relation **FleaMarket**(Artifact, Seller, StandLocation, StandCleanliness, ArtifactOrigin, Price) the following functional dependencies:

- Artifact, Seller → Price
- Artifact → ArtifactOrigin
- Seller → StandLocation, StandCleanliness

What normal form is the relation in?

- Not in 1NF
- ☒ 1NF
- 2NF
- 3NF / BCNF

Question 5 (3 points)

If the relation in Question 4 is not in 3NF / BCNF, normalize the relation to the highest possible normal form (3NF / BCNF) and select the resulting relations. If the relation is in 3NF / BCNF, select the answer “The relation is already in 3NF / BCNF”.

- ☐ The relation is already in 3NF / BCNF.
- ☒ **ArtifactInfo**(Artifact, ArtifactOrigin)
- ☐ **ArtifactInfo**(Artifact, Seller, ArtifactOrigin)
- ☐ **SellerInfo**(Artifact, Seller, StandLocation, StandCleanliness)
- ☒ **SellerInfo**(Seller, StandLocation, StandCleanliness)
- ☐ **FleaMarket**(Artifact, Seller)
- ☒ **FleaMarket**(Artifact, Seller, Price)
- ☐ **FleaMarket**(Artifact, Seller, ArtifactInfo, SellerInfo)
- ☐ **FleaMarket**(Artifact, Seller, Price, ArtifactInfo, SellerInfo)

- ❑ **FleaMarket**(StandLocation, StandCleanliness , ArtifactOrigin, Price)

SQL (40 points)

You are given a dataset that you must import into a DuckDB database on your laptop. Download the file **artifacts_db.sql** from the LearnIT page and copy and paste the SQL statements into DuckDB.

The dataset contains three tables:

The **artifact** table stores information about historical or collectible artifacts. It contains the following columns:

- **id**: Unique identifier for each artifact.
- **description**: Short text describing the artifact.
- **material**: What the artifact is made of (e.g., gold, clay, steel).
- **age**: Age of the artifact in years.

The **buyer** table stores information about buyers who can purchase artifacts. It contains the following columns:

- **cpr_number**: Unique Danish CPR number (format ddmmyyxxxx) for each buyer.
- **firstname**: Buyer's first name.
- **lastname**: Buyer's last name.
- **zip**: Danish 4-digit postal code for the buyer.

Finally, the **sale_history** table tracks the sale of artifacts to buyers over time. It contains the following columns:

- **aid**: ID of the artifact being sold; foreign key referencing artifact(id).
- **cpr_number**: CPR number of the buyer; foreign key referencing buyer(cpr_number).
- **sell_date**: Date and time of the sale.
- **price**: Sale price of the artifact.

Given this dataset, answer the following questions.

Question 6 & 7 (5 points)

How old is the oldest artifact made from gold?

Enter the result and your SQL query that produces exactly this result. SQL queries that output different or additional information will result in a loss of points. SQL queries that are unnecessarily complex will also result in a loss of points. The SQL query must run in DuckDB. SQL queries that do not run in DuckDB will result in a loss of points.

```
-- result: 1500
select max(age)
from artifact
where material = 'Gold';
```

Question 8 & 9 (5 points)

How many buyers have an "m" (case-insensitive) in their firstname or lastname?

Enter the result and your SQL query that produces exactly this result. SQL queries that output different or additional information will result in a loss of points. SQL queries that are unnecessarily complex will also result in a loss of points. The SQL query must run in DuckDB. SQL queries that do not run in DuckDB will result in a loss of points.

```
-- result: 5
select count(*)
from buyer
where firstname ilike '%m%' or lastname ilike '%m%';
```

Question 10 & 11 (5 points)

What is the average selling price of artifacts made of stone?

Enter the result and your SQL query that produces exactly this result. SQL queries that output different or additional information will result in a loss of points. SQL queries that are unnecessarily complex will also result in a loss of points. The SQL query must run in DuckDB. SQL queries that do not run in DuckDB will result in a loss of points.

```
-- result: 1107
select avg(price)
from sale_history
join artifact on sale_history.aid = artifact.id
where material = 'Stone';
```

Question 12 & 13 (5 points)

Who is the last buyer (first name and last name) to have purchased an artifact made of iron? You can assume that no two sales of artifacts happened at the same time.

Select the names from the multiple-choice options and enter your SQL query. SQL queries that output different or additional information will result in a loss of points. SQL queries that are unnecessarily complex will also result in a loss of points. The SQL query must run in DuckDB. SQL queries that do not run in DuckDB will result in a loss of points.

- ☐ Anna Christensen
- ☐ Christian Larsen
- ☐ Emma Jorgensen
- ☐ Ida Sorensen
- ☐ Karla Moller
- ☐ Lars Jensen
- ☐ Marie Andersen
- ☐ Mikkel Madsen
- ☐ Oliver Eriksson
- ☐ Peter Hansen
- ☒ Sofie Pedersen
- ☐ Thomas Nielsen

```
select firstname, lastname
from buyer
join sale_history on buyer.cpr_number = sale_history.cpr_number
join artifact on sale_history.aid = artifact.id
where material = 'Iron'
order by sell_date desc
limit 1;
```

```
-- alternatively:
select firstname, lastname
from buyer
join sale_history on buyer.cpr_number = sale_history.cpr_number
where sell_date = (
    select max(sell_date)
    from sale_history
    join artifact on id = aid
    where material = 'Iron'
);
```

Question 14 & 15 (5 points)

Which buyer(s) (first name and last name) never purchased any artifact?

Select the names from the multiple-choice options and enter your SQL query. SQL queries that output different or additional information will result in a loss of points. SQL queries that are unnecessarily complex will also result in a loss of points. The SQL query must run in DuckDB. SQL queries that do not run in DuckDB will result in a loss of points.

- ☐ Anna Christensen
- ☐ Christian Larsen
- ☐ Emma Jorgensen
- ☐ Ida Sorensen

- ☐ ☒ Karla Moller
- ☐ Lars Jensen
- ☐ Marie Andersen
- ☐ Mikkel Madsen
- ☐ Oliver Eriksson
- ☐ Peter Hansen
- ☐ Sofie Pedersen
- ☐ Thomas Nielsen

```
select firstname, lastname
from buyer
where cpr_number not in (
  select cpr_number
  from sale_history
);
```

```
-- alternatively:
select firstname, lastname
from buyer
left join sale_history on buyer.cpr_number = sale_history.cpr_number
where sale_history.cpr_number is null;
```

Question 16 & 17 (5 points)

Which buyer(s) (first name and last name) have spent the most money purchasing artifacts? Only look at how much they spent and ignore any money they may have earned if someone else bought an artifact that they owned.

Select the names from the multiple-choice options and enter your SQL query. SQL queries that output different or additional information will result in a loss of points. SQL queries that are unnecessarily complex will also result in a loss of points. The SQL query must run in DuckDB. SQL queries that do not run in DuckDB will result in a loss of points.

- ☐ Anna Christensen
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- ☐ ☒ Lars Jensen
- ☐ Marie Andersen
- ☐ Mikkel Madsen
- ☐ Oliver Eriksson
- ☐ ☒ Peter Hansen
- ☐ Sofie Pedersen
- ☐ Thomas Nielsen


```

with spending_per_buyer as (
  select cpr_number, sum(price) as total_spent
  from sale_history
  group by cpr_number
)
select firstname, lastname
from buyer
join spending_per_buyer on buyer.cpr_number = spending_per_buyer.cpr_number
where total_spent = (
  select max(total_spent)
  from spending_per_buyer
);

```

Question 18 & 19 (5 points)

What is the longest gap in days between two sales of the same artifact? Only consider gaps between sales of the same artifact. Ignore any gaps before the first sale or after the last sale of an artifact. Use DuckDB's `date_diff()` function to calculate the difference in days between two dates.

Enter the result and your SQL query that produces exactly this result. SQL queries that output different or additional information will result in a loss of points. SQL queries that are unnecessarily complex will also result in a loss of points. The SQL query must run in DuckDB. SQL queries that do not run in DuckDB will result in a loss of points.

```

-- result: 865
with artifact_sales as (
  select aid, sell_date,
    lag(sell_date) over (partition by aid order by sell_date) as previous_sell_date
  from sale_history
)
select max(date_diff('day', previous_sell_date, sell_date))
from artifact_sales;

```

Question 20 & 21 (5 points)

Which artifact(s) did increase in price for 3 or more consecutive sales? Only consider consecutive sales of the same artifact. Ignore any price changes before the first sale or after the last sale of an artifact.

Select the artifacts from the multiple-choice options and enter your SQL query that produces exactly this result. SQL queries that output different or additional information will result in a loss of points. SQL queries that are unnecessarily complex will also result in a loss of points. The SQL query must run in DuckDB. SQL queries that do not run in DuckDB will result in a loss of points.

☐ Ancient Greek amphora

- ☐ ☒ Byzantine gold coin
- ☐ Celtic bronze torque
- ☐ ☒ Egyptian scarab amulet
- ☐ Medieval ceremonial sword
- ☐ ☒ Mesopotamian clay tablet
- ☐ ☒ Ming Dynasty jade figurine
- ☐ ☒ Persian decorative tile
- ☐ Roman legionary helmet
- ☐ Viking silver brooch

```
with price_changes as (
  select aid, sell_date,
         price - lag(price) over (partition by aid order by sell_date) as price_change
  from sale_history
), three_time_streaks as (
  select aid,
         sum(case when price_change > 0 then 1 else 0 end)
           over (partition by aid order by sell_date
                rows between 2 preceding and current row) as streak
  from price_changes
)
select distinct description
from artifact
join three_time_streaks on artifact.id = three_time_streaks.aid
where streak = 3
order by description;
```

Views (10 points)

Question 22 (3 points)

Create a view named **sale_history_hashed** based on the **sale_history** table. The view must contain the columns **aid**, **cpr_hash**, **sell_date**, and **price**. The columns **aid**, **sell_date**, and **price** must be copies of corresponding columns in the original table. The column **cpr_hash** must contain the SHA256 hash of the original CPR number.

Enter your SQL DDL statement that produces this view. The SQL statement must run in DuckDB. SQL statements that do not run in DuckDB will result in a loss of points.

```
create view sale_history_hashed as
select aid, sha256(cpr_number) as cpr_hash, sell_date, price
from sale_history;
```

Question 23 (3 points)

Create a view named **sale_history_trimmed** based on the **sale_history** table. The view must contain the columns **aid**, **cpr_trim**, **sell_date**, and **price**. The columns **aid**, **sell_date**, and **price** must be copies of corresponding columns in the original table. The column **cpr_trim** must contain only the first four letters of the original CPR number.

Enter your SQL DDL statement that produces this view. The SQL statement must run in DuckDB. SQL statements that do not run in DuckDB will result in a loss of points.

```
create view sale_history_trimmed as
select aid, substr(cpr_number, 1, 4) as cpr_trim, sell_date, price
from sale_history;
```

Question 24 (4 points)

Write down one benefit and one drawback of each view, **sale_history_hashed** and **sale_history_trimmed**.

sale_history_hashed:

- Benefits: CPR numbers are anonymized. Hashes are hard to reverse.
- Drawback: Higher computational cost since hashing is more expensive than simple string operations. Without random salt, hashes are susceptible to brute-force attacks.

sale_history_trimmed:

- Benefits: CPR numbers are anonymized because only part of the number is shown. Fast computation because substringing operations are simple and efficient.
- Drawbacks: Creates many duplicates, leading to information loss and limiting analytical usefulness. Weak privacy protection because date of birth is still shown.

Role-Based Access Control (10 points)

Question 25 (8 points)

In the lecture, you have seen that read and write privileges, such as **select**, **update**, **delete** can be granted to tables. In practice, it is also possible to grant privileges on other object, such as views and functions.

Given the following scenario, write the SQL commands following the Snowflake syntax as shown in Lecture 7. The commands must include the **CREATE ROLE** statements and **GRANT** statements to establish the role hierarchy, assign roles to users, and assign the five privileges shown.

An archaeological research organization manages access to its database using different responsibility levels. There are three main roles within the system: **Researchers**, **Site Leaders**, and **Security**.

Sofie and **Emma** work as researchers. **Sarah** is a site leader responsible for supervising excavation activities at one of the ancient sites. **Mads** and **Carl** are members of the security staff who manage physical access to the sites.

Researchers are responsible for documenting their scientific work. They can view and create new records in the **Study** table when they analyze artifacts and may modify existing study entries if corrections are necessary. However, they are not allowed to delete any study records.

Site leaders oversee the overall research activity at their site but must not see which specific researcher conducted a particular study. For this purpose, they are given access only to a view **StudyView** that displays which artifacts have been studied while hiding the identity of the responsible researcher. Site leaders can read this information but cannot modify it.

Security staff are responsible for checking researchers in and out of the excavation sites. Therefore, they must be able to view and register records in **Visit** table in the system. Since site leaders also supervise daily operations at the site, they inherit the same access rights as security staff and can also read and register visit information.

Each user in the organization is assigned exactly the role that matches their responsibilities, and the permissions of the system reflect these operational duties.

```
create role Researcher;  
create role SiteLeader;  
create role Security;
```

```
grant role Researcher to user Sofie;  
grant role Researcher to user Emma;  
grant role SiteLeader to user Sarah;  
grant role Security to user Mads;  
grant role Security to user Carl;
```

```
grant select, insert, update on table Study to role Researcher;  
grant select on view StudyView to role SiteLeader;  
grant select, insert on table Visit to role Security;
```

```
grant role Security to role SiteLeader;
```

Question 26 (2 points)

Which of the following statements about RBAC are correct?

- ☐ ☒ In RBAC, users can be assigned multiple roles at the same time.
- ☐ ☒ RBAC simplifies permission management in large database systems by grouping privileges into roles.
- ☐ RBAC eliminates the need for authentication mechanisms in database systems
- ☐ RBAC automatically prevents SQL injection attacks.
- ☐ In RBAC, permissions are assigned directly to users and never to roles.

Min/Max Summaries (10 points)

Given a table with the following definition:

```
create table artifact_black_market(name string, price int);
```

The table is split into four row groups with the following min/max summaries:

	Min Name	Max Name	Min Price	Max Price
Row group 1	Bronze Amulet	Iron Dagger	250	600
Row group 2	Clay Tablet	Silver Crown	300	400
Row group 3	Ancient Coin	Jeweled Chalice	150	1000
Row group 4	Marble Statue	Obsidian Spear	400	2000

Question 27 (2 points)

Which row groups must be read at a **minimum** to answer the following query? "Minimum" means that this is the smallest number of row groups necessary to answer the query correctly. Selecting a row group that is not necessary to read will result in a loss of points.

```
select * from artifact_black_market
where name like 'Jeweled%';
```

Select all that apply:

- ☐ Row group 1
- ☐ ☒ Row group 2
- ☐ ☒ Row group 3
- ☐ Row group 4
- ☐ No row groups

Question 28 (2 points)

Which row groups must be read at a **minimum** to answer the following query? "Minimum" means that this is the smallest number of row groups necessary to answer the query correctly. Selecting a row group that is not necessary to read will result in a loss of points.

```
select * from artifact_black_market
where price > 800;
```

Select all that apply:

- ☐ Row group 1
- ☐ Row group 2
- ☐ ☒ Row group 3
- ☐ ☒ Row group 4
- ☐ No row groups

Question 29 (2 points)

Which row groups must be read at a **minimum** to answer the following query? "Minimum" means that this is the smallest number of row groups necessary to answer the query correctly. Selecting a row group that is not necessary to read will result in a loss of points.

```
select * from artifact_black_market
where price = (select max(price) from artifact_black_market);
```

Select all that apply:

- ☐ Row group 1
- ☐ Row group 2
- ☐ Row group 3
- ☐ ☒ Row group 4
- ☐ No row groups

Question 30 (2 points)

Which row groups must be read at a **minimum** to answer the following query? "Minimum" means that this is the smallest number of row groups necessary to answer the query correctly. Selecting a row group that is not necessary to read will result in a loss of points.

```
select * from artifact_black_market
where name <> 'Iron Dagger';
```

Select all that apply:

- ☐ ☒ Row group 1
- ☐ ☒ Row group 2
- ☐ ☒ Row group 3
- ☐ ☒ Row group 4
- ☐ No row groups

Question 31 (2 points)

Which of the following statements about min/max summaries are correct?

Select all that apply:

- ☐ Min/max summaries work best for highly random, uniformly distributed data.
- ☐ ☒ Min/max summaries are particularly effective for columns where data is naturally ordered (e.g., date or timestamp columns).
- ☐ ☒ Min/max summaries allow the database to skip scanning entire partitions if a query predicate falls outside the stored min/max range.
- ☐ Min/max summaries guarantee that no irrelevant rows are ever read during query execution.
- ☐ Min/max summaries replace the need for all indexes in a database system.

BSc Only: OLTP vs. OLAP (5 points)

Question 32 (5 points)

Only answer if you are a BSc student.

From the following database scenarios, select only the scenarios that are examples of **OLAP (Online Analytical Processing)**.

- ☐ A bank processes 10,000 customer transactions per second, including withdrawals and deposits.
- ☐ ☒ A retail company generates a report showing total sales by region for the past 5 years.
- ☐ An airline system books a flight, assigns a seat, and issues a boarding pass.
- ☐ An e-commerce website updates a customer's shopping cart in real-time.
- ☐ ☒ A business analyst creates a dashboard to visualize quarterly revenue trends across departments.
- ☐ A hospital system records patient admissions, discharges, and medication orders.
- ☐ ☒ A marketing team runs a query to identify the top 10% of customers by lifetime value.

- ☐ ☒ A financial institution aggregates data to detect fraudulent transactions over the last month.
- ☐ A university updates student grades and attendance records in real-time.
- ☐ ☒ A supply chain manager analyzes inventory turnover rates for the past fiscal year.

MSc Only: Programming Language Support (5 points)

Question 33 (3 points)

Only answer if you are an MSc student.

Given the following Java program, create a user input that triggers a SQL injection attack.

```
public void updateUserEmail(String email) throws SQLException {  
    // SQL string  
    String sql = "update users set email = '" + email + "' where id = 123";  
  
    // Execute the query  
    try (Statement stmt = conn.createStatement()) {  
        stmt.executeUpdate(sql);  
    }  
}
```

Example solution:

attack@example.com'; delete from users; --

Question 34 (2 points)

Only answer if you are an MSc student.

Explain what the output would be if the attack of your previous answer were executed.

Example solution:

The input would turn the query into:

```
update users set email = 'attack@example.com'; delete from users; -- ' where id = 123
```

The delete from users command is executed, deleting all records from the users table.